

DEAP: A Database for Emotion Analysis using Physiological Signals

Koelstra et al. (2012), IEEE Trans. Affective Computing 3(1) · tags: eeg, eeg_alpha, eeg_beta, calm, stress, arousal, physio

VERBATIM EXCERPT (source: archive-ouverte.unige.ch)

"We present a multimodal dataset for the analysis of human affective states. The electroencephalogram (EEG) and peripheral physiological signals of 32 participants were recorded as each watched 40 one-minute long excerpts of music videos. Participants rated each video in terms of the levels of arousal, valence, like/dislike, dominance and familiarity."

"In this work, we use Russell's valence-arousal scale... emotional states can be placed on a two-dimensional plane with arousal and valence as the horizontal and vertical axes. Arousal ranges from inactive (e.g. uninterested, bored) to active (e.g. alert, excited), whereas valence ranges from unpleasant to pleasant."

Relevance to MedOracle: this is the source dataset for Member 1's physiological branch, and the paper this project's DEAP → 5-class valence/arousal/dominance threshold mapping is built on.